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Thermodynamic assessments of the Sb–La and Sb–Tb systems

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ABSTRACT

In this work, thermodynamic assessments of the Sb–La and Sb–Tb binary systems have been carried out by using the CALPHAD method on the basis of the experimental data including thermodynamic properties and phase equilibria. The Gibbs free energies of the solution phases (liquid, fcc, bcc, hcp, and dhcp) were described by the subregular solution model with the Redlich–Kister equation. The intermetallic compounds, La_2Sb, La_3Sb_2, LaSb, and $LaSb_2$ in the Sb–La system and Sb_2Tb, α-SbTb, β-SbTb, α-Sb_3Tb_4, β-Sb_3Tb_4, and Sb_3Tb_5 in the Sb–Tb system, were treated as stoichiometric phases. A consistent set of thermodynamic parameters has been derived for describing the Gibbs free energy of each solution phase and intermetallic compound in the Sb–La and Sb–Tb systems. A good agreement between the calculated results including phase diagrams and thermodynamic properties and the experimental data in the Sb–La and Sb–Tb systems is obtained.

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1. Introduction

Antimony (Sb) is an important semiconductor material, and the addition of rare earth (RE) elements to semiconductor materials can result in some important changes in the properties [1,2]. Besides, Sb–RE alloys are potential materials owing to their magnetic properties [3]. The phase diagram and thermodynamic properties of the alloy system are of great value in alloy design and processing. Therefore, it is important to understand the phase equilibria in the Sb–RE binary systems.

The CALPHAD method, which is a powerful tool whose use can cut down the cost and time spent during development of materials, effectively provides a clear guideline for material design. In order to design high-performance Sb–RE alloys, it is important to develop the thermodynamic modeling in the Sb–RE binary systems. In the present work, the thermodynamic assessments of the Sb–La and Sb–Tb systems were carried out based on the experimental data of the phase diagrams and thermodynamic properties by means of the CALPHAD method.

2. Experimental information

2.1. The Sb–La system

The evaluation of the available experimental data in the Sb–La system has been done by Massalski [4], where a phase diagram of the Sb–La system was made mainly based on the experimental data from Vogel and Klose's work [5]. There are four intermetallic compounds (La_2Sb, La_3Sb_2, LaSb, and $LaSb_2$) in this binary system: the La_2Sb, LaSb, and $LaSb_2$ phases are formed from the peritectic reactions at about 1460, 1540, and 1110 °C, respectively, whereas the La_3Sb_2 phase melts congruently at about 1690 °C. In view of the fact that the solubility of each intermetallic compound is very limited [4–6], they were treated as stoichiometric compounds in the present work. There are two eutectic reactions, $L \leftrightarrow (\beta\text{-La}) + La_2Sb$ and $L \leftrightarrow (Sb) + LaSb_2$, occurring at 845 and 630 °C. Because the terminal phases in both sides have very narrow homogeneity ranges, they are treated as pure element phases in the present assessment. Due to the lack of the experimental data, the liquidus of the La_3Sb_2 phase is drawn with a dashed line [4]. In addition, the shape of the liquidus described by Massalski [4] is asymmetric at high temperature, which seems to be thermodynamically unlikely. The details of the invariant reaction in this system are shown in Table 1.

No experimental data on the mixing enthalpy of the liquid phase is reported for the Sb–La system. Takeuchi and Inoue [7] predicted the value of the enthalpy of mixing of the liquid phase at 50 at.% La in the Sb–La system; however, no temperature dependence was given in the work of Takeuchi and Inoue [7]. The formation enthalpies and entropies of the intermetallic compounds in the Sb–La system were reported in Refs. [8–10].

2.2. The Sb–Tb system

Based on results of differential thermal analysis (DTA), microscopy, X-ray, chemical, and X-ray fluorescence measurements,

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Table 1
The invariant reactions in the Sb–La system including calculated and experimental results.

Reaction	Composition, at.% Sb			Temperature, °C	Reference
$L \leftrightarrow (\beta\text{-La}) + La_2Sb$	3.0	0	33.3	845	This work
	~3.4	~0	33.3	845	[4]
$L + La_3Sb_2 \leftrightarrow La_2Sb$	25.4	40.0	33.3	1460	This work
	~32.4	40.0	33.3	~1460	[4]
$L + La_3Sb_2 \leftrightarrow LaSb$	52.0	40.0	50.0	1540	This work
	~53.0	40.0	50.0	~1540	[4]
$L + LaSb \leftrightarrow LaSb_2$	73.8	50.0	66.7	1110	This work
	~74.5	50.0	66.7	1110	[4]
$L \leftrightarrow LaSb_2 + (Sb)$	99.6	66.7	100	630	This work
	~99.5	66.7	~100	630	[4]

Table 2
The invariant reactions in the Sb–Tb system including calculated and experimental results.

Reaction	Composition, at.% Tb			Temperature, °C	Reference
$L \leftrightarrow (Sb) + Sb_2Tb$	1.8	0	33.3	623	This work
	~0.8	~0	33.3	623	[13]
$L + (\alpha\text{-SbTb}) \leftrightarrow Sb_2Tb$	4.6	50.0	33.3	740	This work
	~2.2	50.0	33.3	740	[13]
$L + (\alpha\text{-SbTb}) \leftrightarrow (\beta\text{-}Sb_3Tb_4)$	66.8	50.0	57.1	1770	This work
	~61.0	50.0	57.1	1770	[13]
$L + (\beta\text{-}Sb_3Tb_4) \leftrightarrow Sb_3Tb_5$	70.6	57.1	62.5	1650	This work
	~65.1	57.1	62.5	1650	[13]
$L \leftrightarrow (\alpha\text{-Tb}) + Sb_3Tb_5$	87.7	98.3	62.5	1130	This work
	86.0	~99.0	62.5	1130	[13]
$(\beta\text{-Tb}) \leftrightarrow (\alpha\text{-Tb}) + L$	99.1	99.5	93.7	1265	This work
	~98.5	~99.0	~91.6	1265	[13]

Abdusalyamova et al. [11,12] investigated the phase diagram in the Sb–Tb system, and determined the invariant horizontals associated with the inverse peritectic and eutectic reactions as well as the liquidus position. According to the work of Refs. [11, 12], Massalski [13] compiled the phase diagram of the Sb–Tb system. There are four intermetallic compounds (Sb_2Tb, SbTb, Sb_3Tb_4, and Sb_3Tb_5) in this system, in which Sb_3Tb_4 and Sb_3Tb_5 form via peritectic reactions at 1770 ± 15 and 1650 ± 15 °C while SbTb melts congruently at 2160 ± 20 °C. Abdusalyamova et al. [11] detected the existence of polymorphic transformations in Sb_3Tb_4 (α-Sb_3Tb_4 ↔ β-Sb_3Tb_4) and SbTb (α-SbTb ↔ β-SbTb) at 1600 ± 15 and 1860 ± 20 °C, respectively. It should be noted that Abdusalyamova et al. [11] determined the existence of the Sb_2Tb phase occurring at 740 ± 10 °C, and Massalski [13] confirmed that the Sb_2Tb phase forms from the peritectic reaction of $L + \alpha$-SbTb ↔ Sb_2Tb according to Refs. [11,12]. All four intermetallic compounds have very limited solubility and were treated as stoichiometric compounds in Refs. [11,13] and in this work as well. Two eutectic reactions, $L \leftrightarrow (Sb) + Sb_2Tb$ and $L \leftrightarrow (\alpha\text{-Tb}) + Sb_3Tb_5$, occur at about 1.0 at.% Tb and 623 °C and at 86 at.% Tb and 1130 °C. The cooling and heating curves results of DTA measured by Abdusalyamova et al. [11] show that an inverse peritectic reaction occurs at the Tb-rich end of the system at 1265 ± 10 °C and approximately 99 at.% Tb. The melting point of antimony is lowered about 7 °C by small terbium additions, which was interpreted as an indicative of a eutectic reaction at 623 °C. The details of the invariant reaction in this system are shown in Table 2.

No experimental data on the mixing enthalpy of the liquid phase have been reported. Takeuchi and Inoue [7] calculated the mixing enthalpy of the liquid phase at 50 at.% Tb in the Sb–Tb system. The formation enthalpy of the α-SbTb phase was determined by Pratt and Chua [14] at 726 K and the reference states of the Tb and Sb were hcp and A7 structures.

It should be pointed out that the liquidus compositions were obtained from the heating arrests of the high-temperature DTA. Because of the high volatility of antimony at high temperatures, the measured liquidus might have shifted to the Sb-rich side considering the loss of Sb at high temperature. As the Sb-rich alloys were sealed in crucibles, the loss of Sb at high temperature is much less than that of unsealed alloys. These reasons might cause the asymmetry of the liquidus.

3. Thermodynamic models

3.1. Pure elements

Because no absolute values of the Gibbs free energy can be defined, the Gibbs free energy of the pure element i in the ϕ phase, ${}^0G_i^{\phi}(T)$, is referred to the enthalpy for its stable state at 298.15 K, ${}^0H_i^{Ser}(298.15\ \text{K})$. The quantity ${}^0G_i^{\phi}(T) - {}^0H_i^{Ser}(298.15\ \text{K})$ is described as a function of temperature by

$$ {}^0G_i^{\phi}(T) - {}^0H_i^{Ser}(298.15\ \text{K}) = a + bT + cT\ln(T) + dT^{-2} + eT^3 + fT^{-1} + gT^7 + hT^{-9}. \quad (1) $$

Eq. (1) for each element in the present work is taken from the SGTE database [15]. The (Sb) and (α-La) terminal phases are described as the pure component phase in the present work.

3.2. Solution phases

The Gibbs free energies of the liquid, fcc (β-La), bcc ((β-Tb) and (γ-La)), and hcp (α-Tb) phases in the Sb–La and Sb–Tb binary systems are described by the subregular solution model with the Redlich–Kister formula [16]. Taking the A–B system as an example, the Gibbs free energies of the solution phases are described by the following expression:

$$ G_m^{\phi} = {}^0G_A^{\phi}x_A + {}^0G_B^{\phi}x_B + RT(x_A\ln x_A + x_B\ln x_B) + {}^EG^{\phi}, \quad (2) $$

where G_i^{ϕ} is the molar Gibbs free energy of pure element i in the respective reference state with the ϕ phase, which is taken from

Table 3
Thermodynamic parameters in the Sb–La and Sb–Tb systems optimized in the present work.

System	Phase	Thermodynamic parameters
Sb–La	Liquid phase	${}^{0}L^{\text{liquid}}_{\text{La,Sb}} = -290000 - 39T$
		${}^{1}L^{\text{liquid}}_{\text{La,Sb}} = 4500 - 42T$
		${}^{2}L^{\text{liquid}}_{\text{La,Sb}} = 59000$
	La_2Sb phase	$G^{\text{La}_2\text{Sb}}_{\text{La:Sb}} - 2G^{\text{ser}}_{\text{La}} - G^{\text{ser}}_{\text{Sb}} = -396100 + 45.94T$
	La_3Sb_2 phase	$G^{\text{La}_3\text{Sb}_2}_{\text{La:Sb}} - 3G^{\text{ser}}_{\text{La}} - 2G^{\text{ser}}_{\text{Sb}} = -697800 + 58.15T$
	La_2Sb phase	$G^{\text{LaSb}}_{\text{La:Sb}} - G^{\text{ser}}_{\text{La}} - G^{\text{ser}}_{\text{Sb}} = -274000 + 24.94T$
	$LaSb_2$ phase	$G^{\text{LaSb}_2}_{\text{La:Sb}} - G^{\text{ser}}_{\text{La}} - 2G^{\text{ser}}_{\text{Sb}} = -289800 + 19.14T$
Sb–Tb	Liquid phase	${}^{0}L^{\text{liquid}}_{\text{Sb,Tb}} = -250300 - 5T$
		${}^{1}L^{\text{liquid}}_{\text{Sb,Tb}} = -23600 + 40T$
		${}^{2}L^{\text{liquid}}_{\text{Sb,Tb}} = 81200 - 13T$
		${}^{3}L^{\text{liquid}}_{\text{Sb,Tb}} = -29000$
	bcc phase	${}^{0}L^{\text{bcc}}_{\text{Sb,Tb}} = -202740$
	hcp phase	${}^{0}L^{\text{hcp}}_{\text{Sb,Tb}} = -196800$
	Sb_2Tb phase	$G^{\text{Sb}_2\text{Tb}}_{\text{Sb:Tb}} - 2G^{\text{ser}}_{\text{Sb}} - G^{\text{ser}}_{\text{Tb}} = -251780 + 31.59T$
	α-SbTb phase	$G^{\alpha\text{-SbTb}}_{\text{Sb:Tb}} - G^{\text{ser}}_{\text{Sb}} - G^{\text{ser}}_{\text{Tb}} = -229036.5 + 12.06T$
	β-SbTb phase	$G^{\beta\text{-SbTb}}_{\text{Sb:Tb}} - G^{\text{ser}}_{\text{Sb}} - G^{\text{ser}}_{\text{Tb}} = -227970 + 11.56T$
	α-Sb_3Tb_4 phase	$G^{\alpha\text{-Sb}_3\text{Tb}_4}_{\text{Sb:Tb}} - 3G^{\text{ser}}_{\text{Sb}} - 4G^{\text{ser}}_{\text{Tb}} = -735936.5 + 39.34T$
	β-Sb_3Tb_4 phase	$G^{\beta\text{-Sb}_3\text{Tb}_4}_{\text{Sb:Tb}} - 3G^{\text{ser}}_{\text{Sb}} - 4G^{\text{ser}}_{\text{Tb}} = -735000 + 38.84T$
	Sb_3Tb_5 phase	$G^{\text{Sb}_3\text{Tb}_5}_{\text{Sb:Tb}} - 3G^{\text{ser}}_{\text{Sb}} - 5G^{\text{ser}}_{\text{Tb}} = -760798 + 36.33T$

the compilation by Dinsdale [15], and the term ${}^{E}G^{\phi}$ denotes the excess Gibbs free energy, which is expressed by Redlich–Kister polynomial [16] as

$$
\begin{aligned}
{}^{E}G^{\phi} &= {}^{0}L^{\phi}_{AB}x_Ax_B + {}^{1}L^{\phi}_{AB}(x_A - x_B)x_Ax_B + {}^{2}L^{\phi}_{AB}(x_A - x_B)^2x_Ax_B + \cdots \\
&= \sum_{m=0}^{n} {}^{m}L^{\phi}_{AB}(x_A - x_B)^m x_Ax_B, \qquad (3)
\end{aligned}
$$

where ${}^{m}L^{\phi}_{AB}$ are the interaction parameters between A and B atoms, expressed as

$$
{}^{m}L^{\phi}_{AB} = a + bT. \qquad (4)
$$

In this work, the parameters a and b are evaluated based on the experimental data.

3.3. Stoichiometric compounds

All the intermetallic compounds in the Sb–La and Sb–Tb systems are treated as stoichiometric phases. The Gibbs free energy per mole of formula unit A_mB_n can be expressed as follows:

$$
\Delta G^{A_mB_n}_f = G^{A_mB_n}_f - mG^{\text{Ser}}_A - nG^{\text{Ser}}_B = a' + b'T, \qquad (5)
$$

where $\Delta G^{A_mB_n}_f$ denotes the standard Gibbs free energy of formation of the stoichiometric compound from the pure elements. The term G^{Ser}_i is the molar Gibbs free energy of pure element i ($i = A, B$ respectively) in its standard reference state. The parameters a' and b' are evaluated in the present work.

4. Optimization results and discussion

Most of the above experimental information was selected for the evaluation of the thermodynamic model parameters. The optimization was carried out using the Thermo-Calc software developed by Sundman et al. [17]. The thermodynamic parameters of all phases in these two systems were obtained, and they are listed in Table 3.

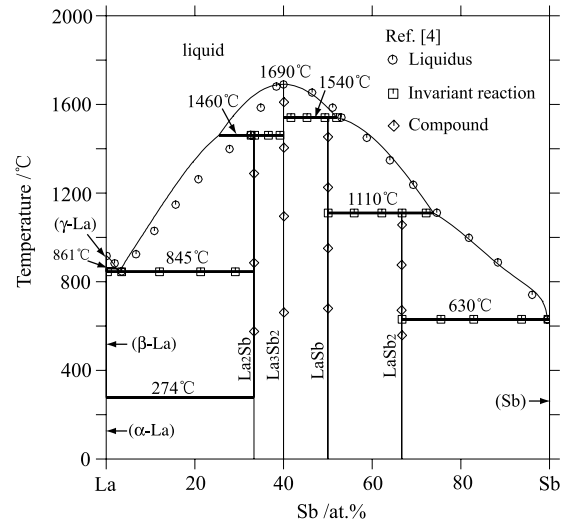

Fig. 1. The calculated phase diagram of the Sb–La system together with the data. *Source:* Taken from [4].

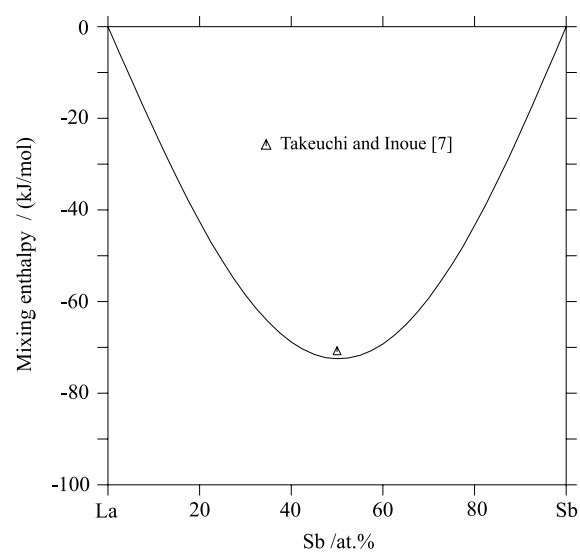

Fig. 2. The calculated mixing enthalpies of the liquid phase at 2000 K in the Sb–La system together with the results reported by Takeuchi and Inoue [7]. *Source:* Reference states: Sb (liquid) and La (liquid).

4.1. The Sb–La system

Fig. 1 shows the calculated phase diagram of the Sb–La system together with the data taken from [4]. It can be seen that the calculated phase diagram is in reasonable agreement with the experimental data except for the liquidus of the La_3Sb_2 phase in the La-rich side. According to the results reported by Okamoto and Massalski [18,19], the liquidus drawn by dashed line in Ref. [4] seems to be impossible in thermodynamics. The reason has been mentioned before, and has also been pointed out by Li et al. [20]. Thus, the liquidus of the La_3Sb_2 phase calculated in the present work is more symmetric and realistic in terms of thermodynamic behavior. As a result, the deviations between the calculated and the experimental results are reasonably acceptable. The calculated peritectic temperatures of the intermetallic compounds (La_2Sb, LaSb, and $LaSb_2$) are 1460, 1540, and 1110 °C, respectively, and the calculated congruent melting temperature of La_3Sb_2 is 1690 °C; these values are in good agreement with the experimental data [4]. The calculated eutectic points of the eutectic reactions, $L \leftrightarrow (\beta\text{-La}) + La_2Sb$ and $L \leftrightarrow (\text{Sb}) + LaSb_2$, are 3.0 at.% Sb at 845 °C

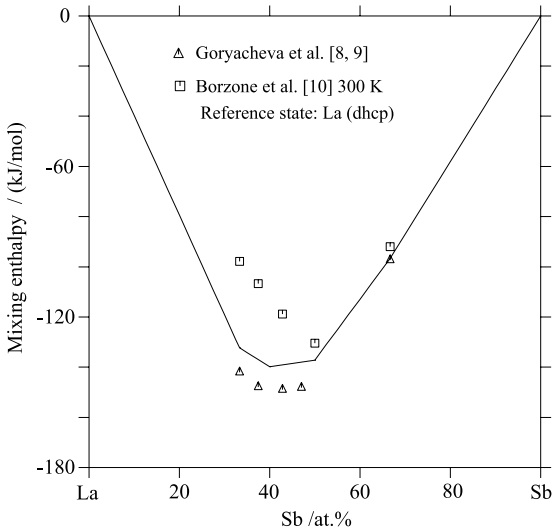

Fig. 3. The calculated formation enthalpies of solid alloys at 730 K in the Sb–La system together with the experimental data [8–10].
Source: Reference states: Sb (Rho_A7) and La (fcc).

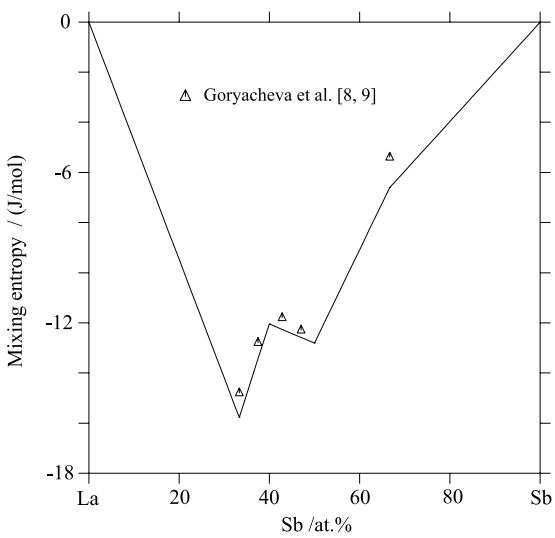

Fig. 4. The calculated formation entropies of solid alloys at 730 K in the Sb–La system together with the experimental data [8,9].
Source: Reference states: Sb (Rho_A7) and La (fcc).

and 99.6 at.% Sb at 630 °C, which agree well with the results taken from [4]. The details of the invariant reactions in this binary system together with the data from the literature are shown in Table 1.

Fig. 2 presents the calculated mixing enthalpies of the liquid phase at 2000 K, together with the results reported by Takeuchi and Inoue [7], from which it can be seen that a good agreement is obtained. Figs. 3 and 4 represent the calculated formation enthalpies and formation entropies of solid alloys at 730 K together with the experimental data [8–10]. It can be seen from Fig. 4 that the calculated curve agrees very well with the results of Goryacheva et al. [8,9]. It was found during the optimization process that better agreement with the data of Borzone et al. [10] could not be achieved whilst maintaining overall agreement with other datasets.

4.2. The Sb–Tb system

The calculated phase diagram of the Sb–Tb system together with the data taken from Ref. [13] is shown in Fig. 5, from which it

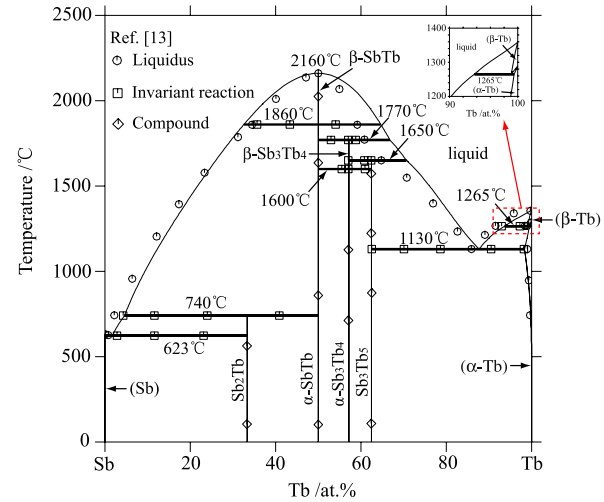

Fig. 5. The calculated phase diagram of the Sb–Tb system together with the data (symbols) obtained from [13].

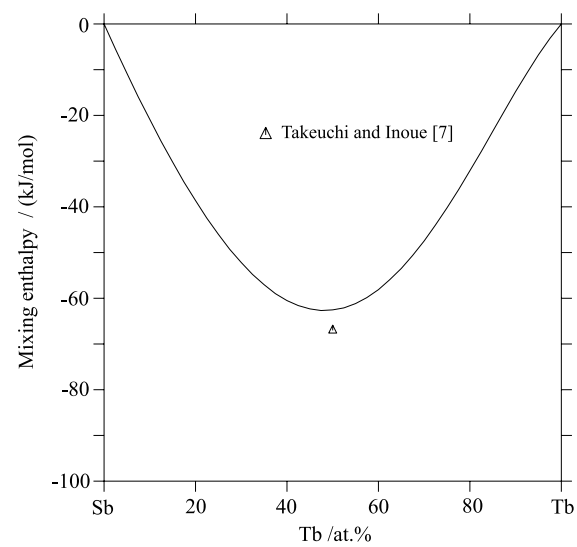

Fig. 6. The calculated mixing enthalpies of the liquid phase at 2000 K in the Sb–Tb system together with the results reported by Takeuchi and Inoue [7].
Source: Reference states: Sb (liquid) and Tb (liquid).

can be seen that the calculated phase diagram is in good agreement with the experimental data. The calculated liquidus line is different from that shown in review paper [13], where the liquidus of the SbTb phase is asymmetric. However, it is unusual considering the thermodynamic behavior [18,19]. This is similar to the Sb–La system. Therefore, the calculated liquidus of the SbTb phase is reasonably acceptable in terms of thermodynamic behavior. The congruent melting temperature of SbTb calculated in the present work is 2160 °C, in agreement with the experimental result [11]. The calculated peritectic temperatures of the three intermetallic compounds (Sb_2Tb, Sb_3Tb_4, and Sb_3Tb_5) are 740, 1770, and 1650 °C, respectively, and the calculated eutectic temperatures in this binary system are 620 and 1130 °C. The calculated temperature of the inverse peritectic reaction in the Tb-rich side, (β-Tb) $\leftrightarrow$ L + (α-Tb), is 1265 °C. All calculated temperatures are in good agreement with the experimental data [13] and are shown in Table 2.

The calculated mixing enthalpies of the liquid phase at 2000 K together with the results reported by Takeuchi and Inoue [7] are shown in Fig. 6, from which it can be seen that a good agreement is obtained. Fig. 7 shows the calculated formation enthalpies of solid

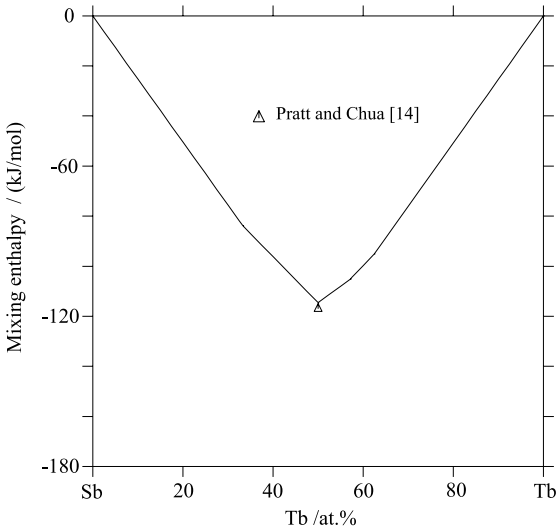

Fig. 7. The calculated formation enthalpies of solid alloys at 726 K in the Sb–Tb system together with the experimental data [14].
Source: Reference states: Sb (Rho_A7) and Tb (hcp).

alloys at 726 K in this binary system together with the experimental data [14]. A reasonable agreement is observed between the calculated values and values taken from the literature.

5. Conclusion

The phase diagrams and thermodynamic properties of the Sb–La and Sb–Tb systems were thermodynamically assessed on the basis of the experimental data available in the literature. A consistent set of thermodynamic parameters for various thermodynamic models has been obtained, and the calculated peritectic and congruent melting temperatures of the intermetallic compounds in these two systems are in good agreement with the experimental data. The calculated liquidus lines in the Sb–La and Sb–Tb binary systems are more reasonable than those reported in the literature in terms of the thermodynamic behavior, and the calculated phase equilibria and thermodynamic properties agree well with most of the literature data.

Acknowledgments

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Appendix. Supplementary data

Supplementary material related to this article can be found online at http://dx.doi.org/10.1016/j.calphad.2012.01.002.

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